

# Jude Anthone R. Duarte

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## EXPERIENCE

### FlyRank AI

June 2026 – Present

Backend AI Engineering Intern

Remote

- Developed backend applications using TypeScript, Node.js, and Express.js, building RESTful APIs, CRUD operations, and modular backend services.
- Integrated PostgreSQL into Dockerized backend applications using a repository-based architecture, enabling persistent and scalable data storage.
- Implemented secure authentication using Supabase Auth, JWT, and middleware, protecting API endpoints and documenting them with Swagger/OpenAPI.
- Built backend data pipelines using web scraping and asynchronous background jobs, preparing structured data for AI-driven applications.

## PROJECTS

### BrewCrafter Smart Café Management System | *React, Javascript, Express.js, Node.js, TailwindCSS, PostgreSQL*

- Built a full-stack web-based café management system for ordering and business operations.
- Implemented a custom drink builder with dynamic pricing and ingredient selection.
- Developed inventory management, order tracking, and sales analytics modules.
- Created RESTful APIs and integrated PostgreSQL for data persistence and system functionality.

### Neural Cellular Automata-Based Random Number Generator for Blockchain Financial Transparency System | *Python, PyTorch, Stable-Baselines3 (PPO), React, Node.js, Express.js, PostgreSQL, SHA-256, ECDSA*

- Developed a machine learning-based random number generator (NCA-RNG) using Neural Cellular Automata to produce cryptographic bitstreams for blockchain applications.
- Trained and optimized the model using Proximal Policy Optimization (PPO) with statistical reward metrics including Shannon Entropy, Autocorrelation, and NIST SP 800-22 tests.
- Built a blockchain-based financial transparency system for student organizations with transaction tracking, auditability, and role-based access control.

### EACON Smart Blogging Web Application | *React, TypeScript, Node.js, Express.js, TailwindCSS, PostgreSQL*

- Built a full-stack blogging and knowledge-sharing platform with posts, learning materials, comments, likes, shares, tagging, and role-based access control (Student, Faculty, Admin).
- Developed an admin analytics dashboard for tracking user activity, content engagement, downloads, and system-wide performance metrics.
- Implemented secure JWT-based authentication with session persistence and role-based authorization.
- Designed a layered backend architecture (controllers, services, middleware) and integrated RESTful APIs with file uploads, validation, and rate limiting.

### ECS - Information Management System for EARIST | *React, Next.js, Go, PostgreSQL, TailwindCSS*

- Developed a full-stack web-based information management system for EARIST Extension Services to handle project requests, monitoring, and documentation.
- Implemented role-based access control system supporting System Administrator, Program Chair, Project Head, and Public User roles with secure authentication and authorization.
- Designed and developed core system modules including project request submission, proposal management, document processing workflow, and status tracking.

## TECHNICAL SKILLS

**Languages:** JavaScript, TypeScript, Python, Java, C++, SQL (PostgreSQL, MySQL), HTML, CSS

**Framework & Technologies:** React.js, Node.js, Express.js, Tailwind CSS, RESTful APIs, Authentication & Authorization

**Developer Tools:** Git, GitHub, VS Code, Postman, Eclipse

**Data Science & AI:** Python, Pandas, NumPy, Matplotlib, Scikit-Learn

## EDUCATION

Eulogio "Amang" Rodriguez Institute of Science and Technology - Manila

Manila City, Philippines

*Bachelor of Science in Computer Science*

2023 - 2027

PHINMA-Saint Jude College – Manila

Manila City, Philippines

*Information and Communications Technology*

2021 - 2023